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PINNGraPE: Physics-Informed Neural Network for Gravitational wave Parameter Estimation

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GR24-Amaldi16
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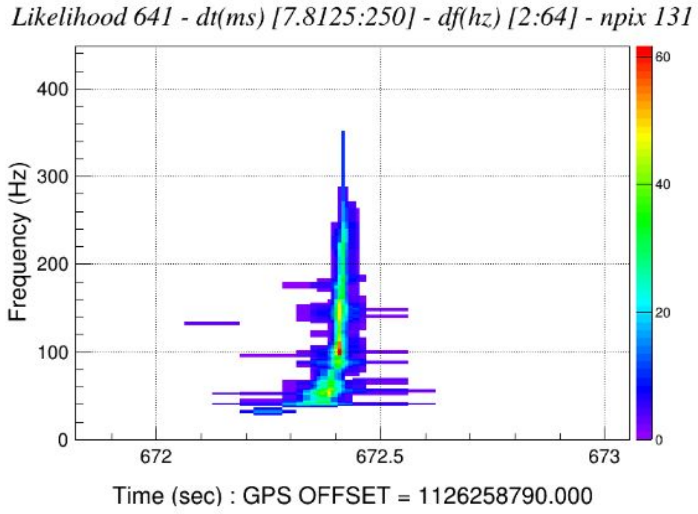
Coherent WaveBurst (cWB)



- Makes detections based upon excess coherent energy across a network of detectors
  - Transforms data into the time-frequency (TF) domain
  - Maximises likelihood over 7 TF resolutions
- Often used to search for short transient events which are not well modeled, however is also capable of detecting compact binary coalescences (CBCs)
  - Doesn't use modeled waveforms in the detection of CBCs
  - Source parameter estimation (PE) is not optimal

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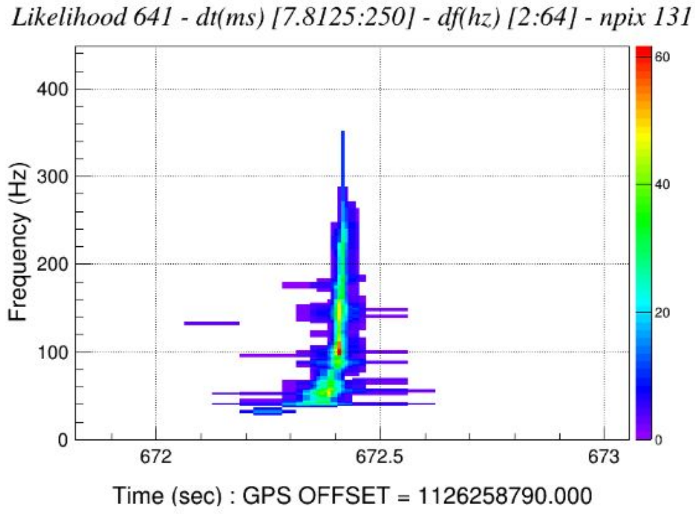

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motivation!

Physics Informed Neural Networks (PINNs)

- Neural networks used to solve problems governed by physical laws
- Eg. solve differential equations, exploit boundary conditions
  - Information contained in the loss function in the form of differential equations or physical relations

PINNGraPE

Aim: use a PINN to infer the mass properties of binary black holes (BBH) detected by cWB

- Component masses: m_1, m_2
- Total mass: $M_{tot} = m_1 + m_2$
- Symmetric mass ratio: $\eta = \frac{m_1 m_2}{(m_1 + m_2)^2}$
- Chirp mass: $\mathcal{M} = M_{tot}\eta^{3/5}$

Describes mass distribution with the binary system. Maximum of 0.25 for $m_1 = m_2$

Combination of masses which describes the frequency evolution of the signal

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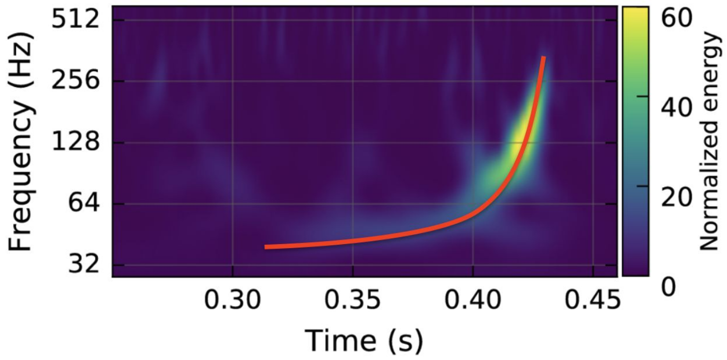

Use the 1.5 post-newtonian (PN) expansion of PDE describing the chirp of a BBH signal:

$$\frac{df}{dt} = \underbrace{\frac{96}{5}\pi^{8/3}\left(\frac{GM_\odot}{c^3}\frac{M_{\rm tot}}{M_\odot}\eta^{3/5}\right)^{5/3} f^{11/3}}_{\text{Newtonian}} \underbrace{\left[1-\left(\frac{743}{336}+\frac{11}{4}\eta\right)\varepsilon^{2/3}+4\pi\varepsilon\right]}_{\text{1.5PN expansion}}$$

$$\varepsilon = \frac{GM_\odot}{c^3}\left(\frac{M_{\rm tot}}{M_\odot}\right)\pi f$$

C. Cutler and E. Flanagan 1994, Phys. Rev. D 49, 26584

PINNGraPE

Aim: use a PINN to infer the mass properties of binary black holes (BBH) detected by cWB

How to get this information from cWB?

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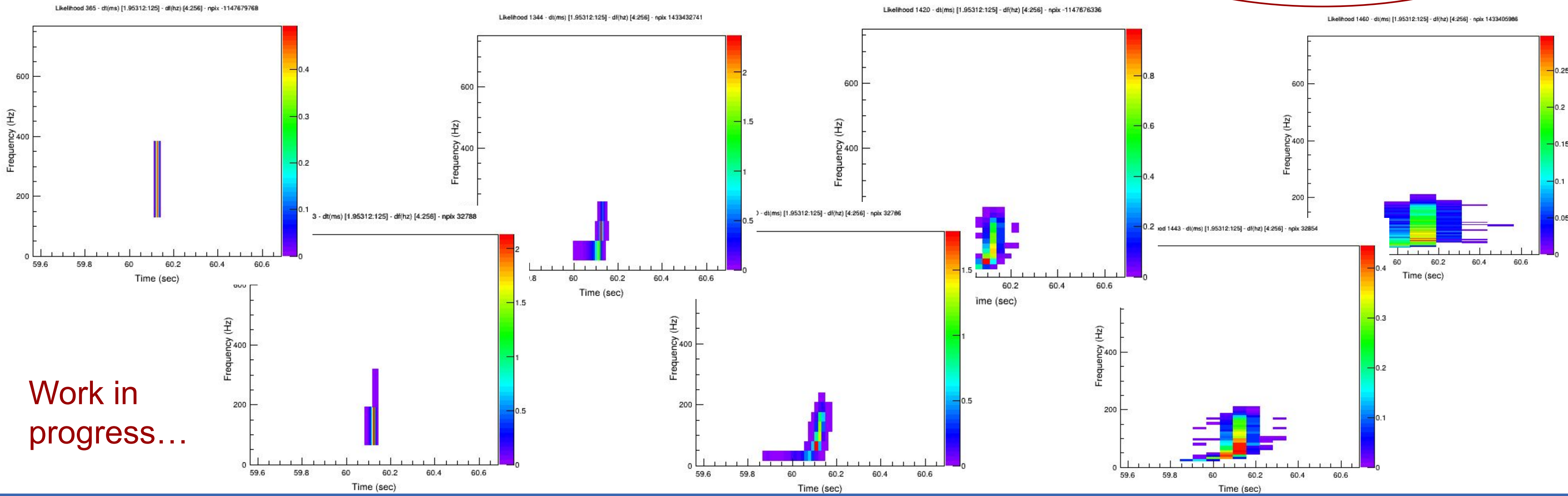

Work in progress…

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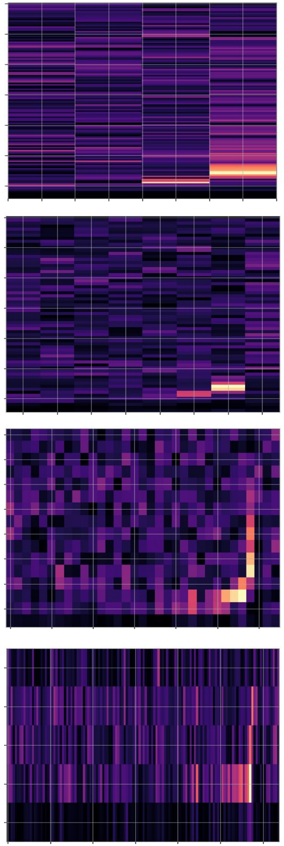
Data - PyCBC injections

Create injections which 'mimic' outputs from cWB

- Produce TF spectrograms over 7 resolutions
- Injected into Gaussian noise coloured with aLIGO PSD ⟶ DCC:T1800044-v5
- Random η and M_{tot} generated over uniform distribution
  - Calculate corresponding $\mathcal{M}$, m_1, m_2

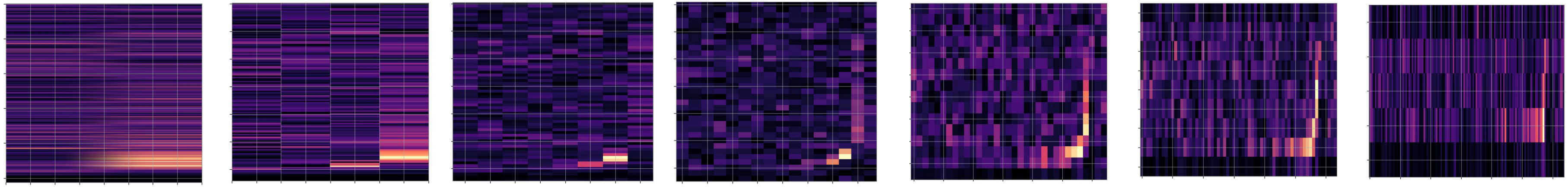
$$\mathcal{M} = M_{tot}\eta^{3/5} \qquad m_1 = \frac{M_{tot}}{2}(1 + \sqrt{1 - 4\eta}) \qquad m_2 = \frac{M_{tot}}{2}(1 - \sqrt{1 - 4\eta})$$

- Numerical integration used to find f(t) ⟶ calculate df/dt



Data - PyCBC injections

- Injections parameters:
    - Total mass [40,100] $M_\odot$
    - η [0.1,0.25]
    - distance [500,1000] Mpc
- 10,000 injections: 70% training, 20% validation, 10% testing
- Each injection has 7 resolutions:
    - (2,257), (4,129), (8,65), (16,33), (33,16), (64,9), (128,5)



Multi-branch CNN

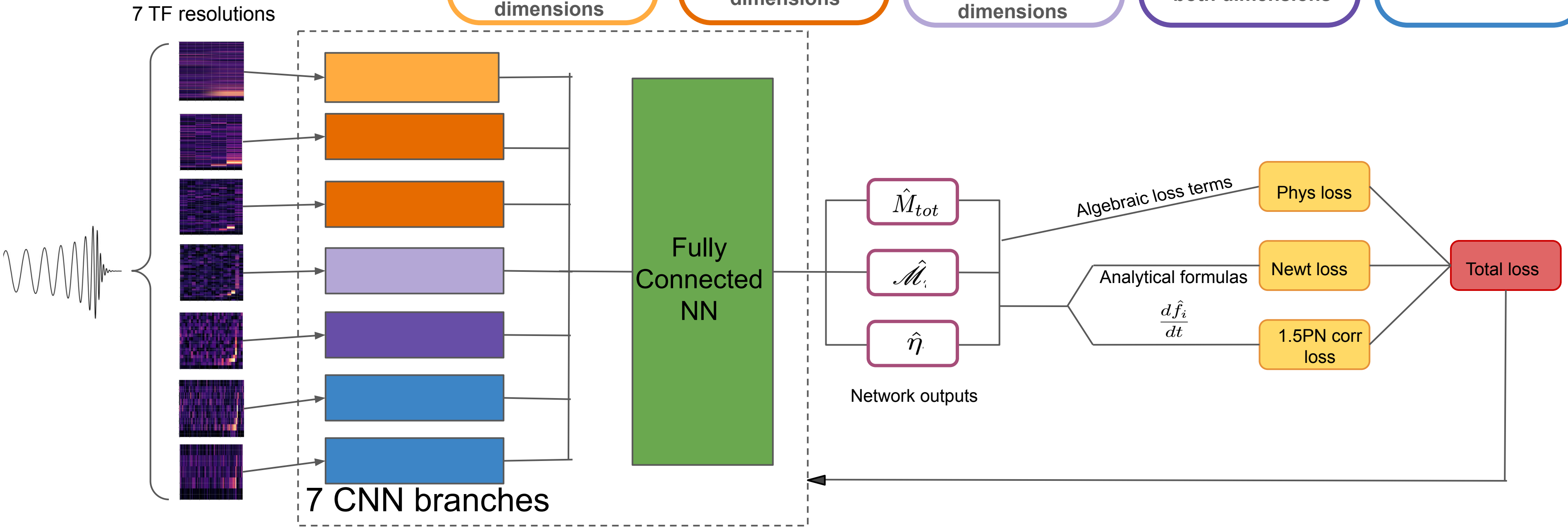
Defined to handle **high F, low t** resolutions
MaxPool2D quarters, then halves **frequency dimensions**

Defined to handle **high F, low t** resolutions
MaxPool2D halves **frequency dimensions**

Defined to handle **almost balanced F, t** resolutions
MaxPool2D halves **frequency**, then halves **both dimensions**

Defined to handle **almost balanced F, t** resolutions
MaxPool2D halves **time**, then halves **both dimensions**

Defined to handle **low F, high t** resolutions
MaxPool2D halves **time dimensions**



Physics-informed loss

$$\boxed{\mathcal{M} = M_{tot}\eta^{3/5}}$$

$$\mathcal{L} = \mathcal{L}_{Newt} + \mathcal{L}_{1.5PN} + \mathcal{L}_{Phys} = \boxed{\frac{\beta_1}{B}\sum_{i=1}^{B}\left(\frac{d\hat{f}_i}{dt} - \frac{df_i}{dt}\right)^2_{Newt}} + \boxed{\frac{\beta_2}{B}\sum_{i=1}^{B}\left(\frac{d\hat{f}_i}{dt} - \frac{df_i}{dt}\right)^2_{1.5PN-Newt}}$$

$$\boxed{+\frac{\beta_3}{B}\sum_{i=1}^{B}\left(\hat{M}_{tot,i}\hat{\eta}_i^{3/5} - \hat{\mathcal{M}}_i\right)^2 + \frac{\beta_4}{B}\sum_{i=1}^{B}\left(\hat{\mathcal{M}}_i\hat{\eta}_i^{-3/5} - \hat{M}_{tot,i}\right)^2 + \frac{\beta_5}{B}\sum_{i=1}^{B}\left(\left(\frac{\hat{\mathcal{M}}_i}{\hat{M}_{tot,i}}\right)^{3/5} - \hat{\eta}_i\right)^2}$$

df/dt terms split due to orders of magnitude difference in values

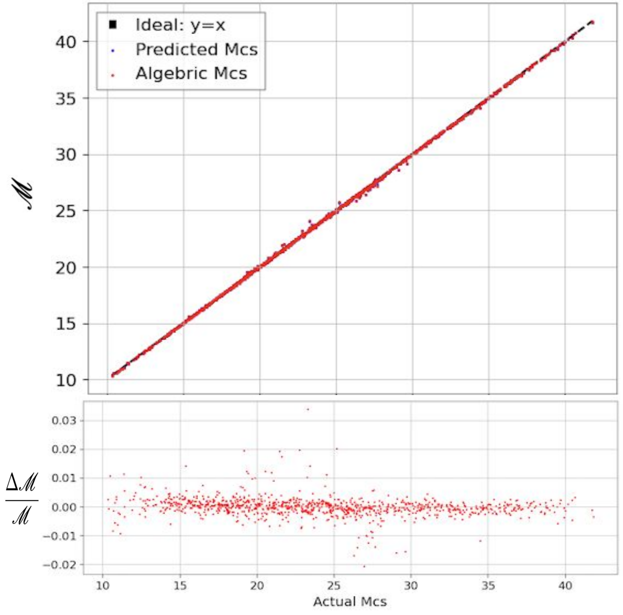
Physical algebraic terms avoid direct output vs. target terms

- Exploits physical redundancy between $\mathcal{M}$, M_{tot} & η , helping to enforce physical boundaries

Results

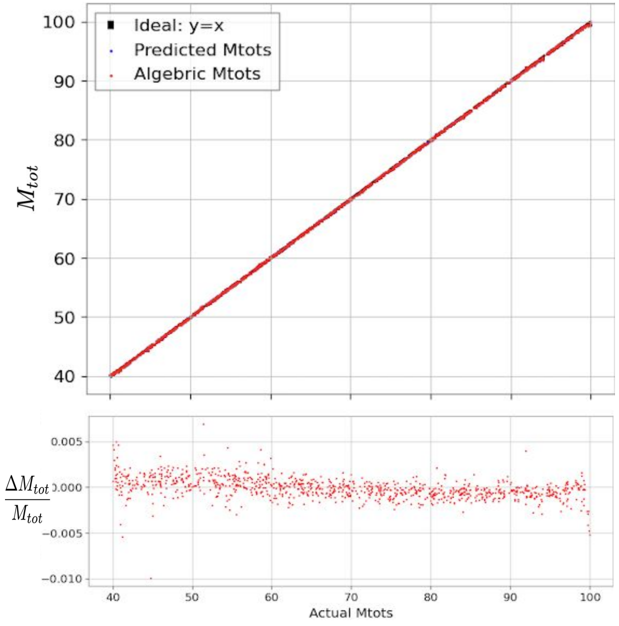
Time required (GPU):
Training: 1h20min
Testing: ~2ms per signal

$\mathscr{M}$



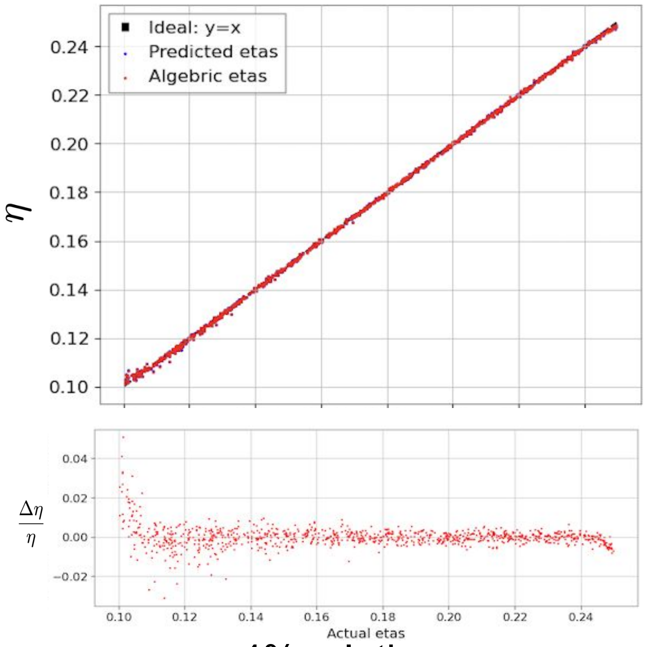
3% relative error

M_{tot}



0.5% relative error

η

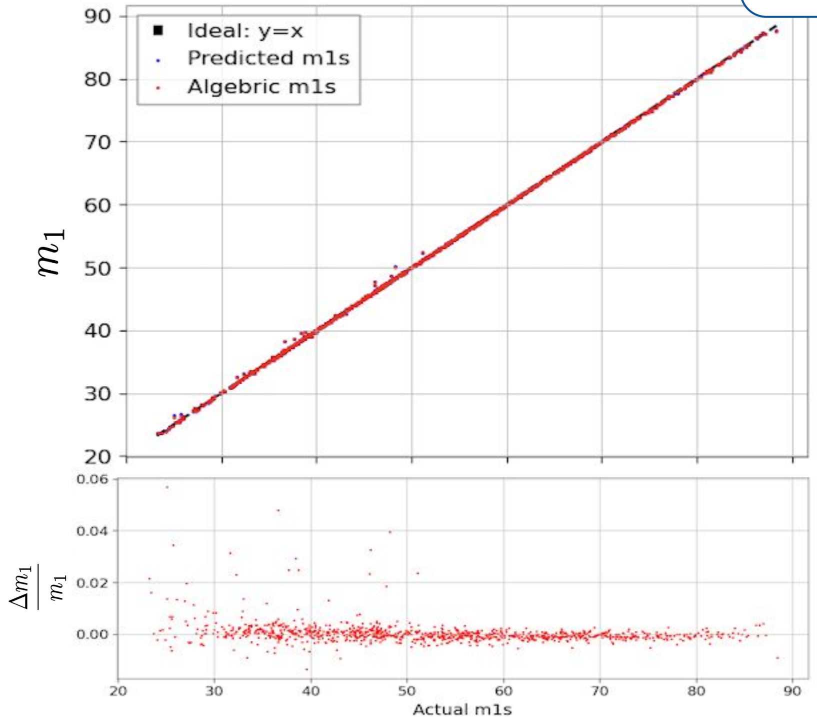


4% relative error

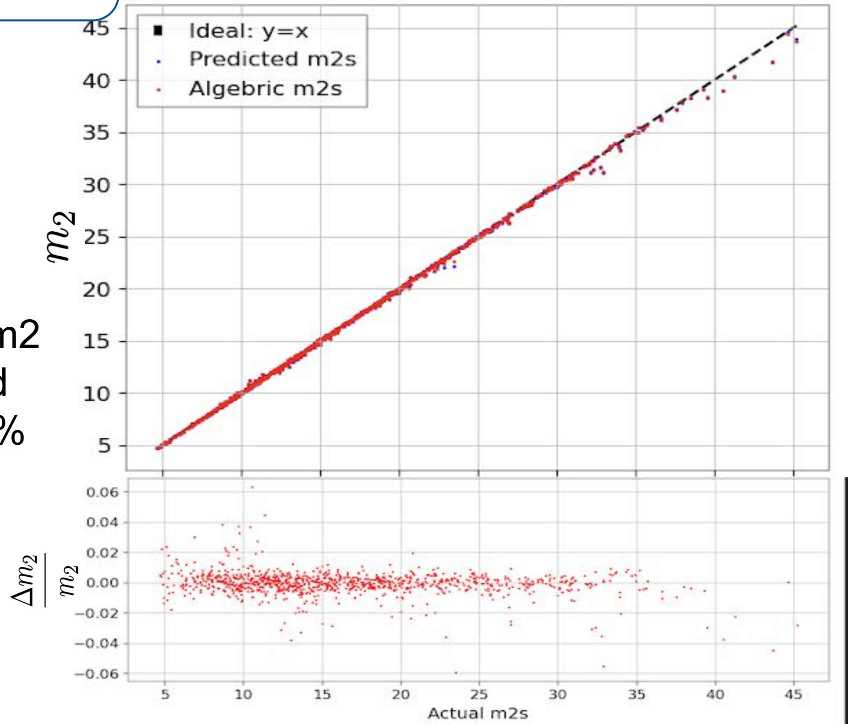
Results

Time required (GPU):
Training: 1h20min
Testing: ~2ms per signal

m_1



m_2



m1 and m2 inferred within 6% error

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Conclusions and future steps

- PINN takes time frequency representations of a signal and infers mass properties
- Able to infer mass properties of BBH to within 6% error

- Future steps:
  - Implement data directly from cWB
  - Test on instances of transient noise (glitches)
  - Test on increased range of masses
  - Increase to further post-newtonian terms to increase the number of parameters inferred
  - Further investigations into model architecture

Acknowledgements

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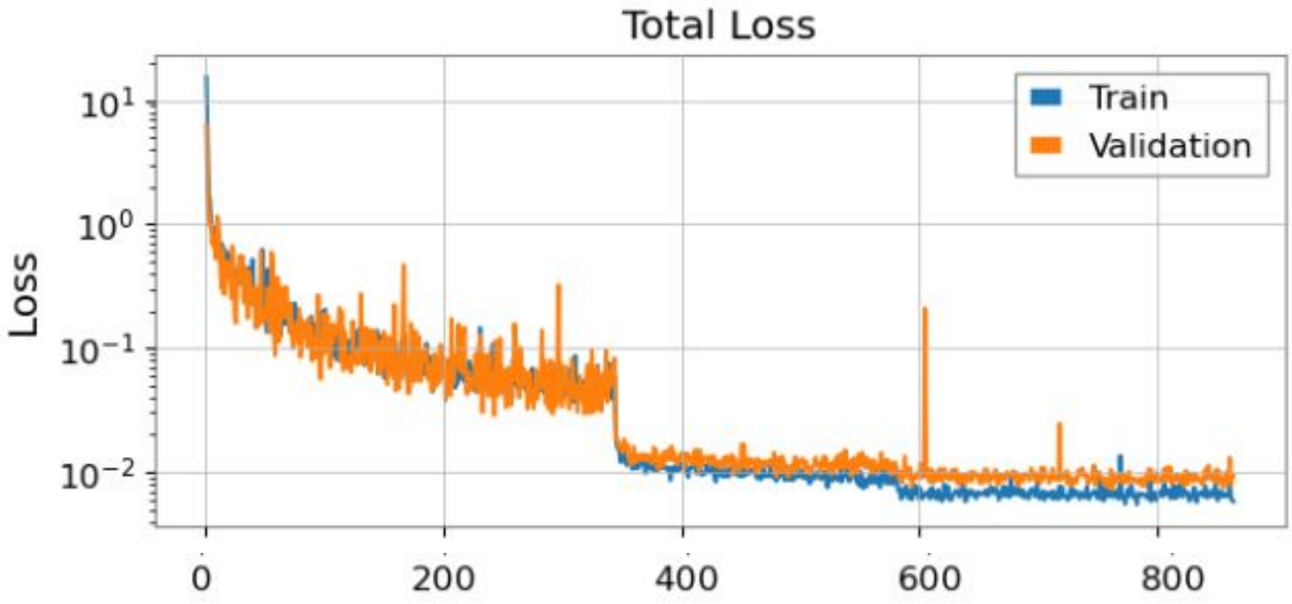
Run hyperparameters

- Loss factors: β_1 = 1e-6, β_2 = 1e4, β_3 = β_4 = β_5=1
- Optimizer: AdamW
- Scheduler: ReduceLROnPlateau (factor=0.1, patience =100)
- Break for learning rate <1e-6

- Final train loss: 5.7e-3
- Final valid loss: 9.2e-3
- Time required (GPU) for training: 1h20min
- Testing: ~2ms per signal

Losses


Total Loss
Train
Validation
Loss
10^{1}
10^{0}
10^{-1}
10^{-2}
0
200
400
600
800